Xin Yao

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Research Interests

My research lies at the intersection of machine learning, wireless communication systems, edge computing, and information theory. I combine theoretical analysis with algorithm and system design to develop efficient, secure, and adaptive methods for distributed learning and dynamic network control, and I evaluate them on real-world wireless testbeds that I build from the physical layer up, from low-power LoRa networks to USRP software-defined radios running srsRAN 4G/5G and protocols I implement. SONAR, submitted to ICC 2027, is where I began bringing AI into this work, with language-model agents that generate network controllers tested on physical radios.

Keywords. Reinforcement learning for wireless networks; Lyapunov optimization under real-world constraints; LoRa and low-power multi-hop networks; software-defined radio and physical-layer implementation; wireless testbeds and remote experimentation platforms; language-model agents for network-controller synthesis; decentralized and federated learning; error feedback and communication compression; information-theoretic secure aggregation; mmWave and RIS-assisted systems; wireless sensing and indoor localization.

Current research topics and collaborating institutions

- Lyapunov-based control and path optimization for resilient multi-hop LoRa networks under interference, outages, and jamming (with Idaho National Laboratory)
- Multi-agent reinforcement learning for spectrum sharing and power allocation, evaluated on live radios and measured channel data (with Idaho National Laboratory and Technische Universität Berlin)
- Communication-efficient decentralized learning with compression and error feedback (with the U.S. Army DEVCOM Army Research Laboratory and the University of Exeter)
- Information-theoretic secure aggregation for federated learning (with Huazhong University of Science and Technology, the University of North Texas, and Technische Universität Berlin)
- Language-model agents that synthesize wireless network controllers with hardware in the loop (SONAR)
- Device-free wireless sensing and indoor localization with commodity LoRa radios
- Shared remote-access infrastructure for NextG and IoT testbeds (NSF-funded UnionLabs, with Virginia Tech, the University of Miami, Ajou University, Georgia Southern University, and The Ohio State University)

Research Statement (Summary)

The long-term goal of my research is to make learning-driven wireless networks practical, meaning networks that use machine learning to allocate resources, share spectrum, and adapt the physical layer, with guarantees of convergence, privacy, and reliability that continue to hold after deployment on real hardware. A central motivation is the gap between simulation and reality. Many learning algorithms proposed for wireless systems perform well in simulation but degrade on real hardware, where carrier-frequency offsets, timing errors, interference, and hardware outages violate their design assumptions. I address this gap from both directions, by developing algorithms whose analysis accounts for real-world constraints and by building experimental infrastructure that makes over-the-air validation systematic.

My doctoral research follows four threads. (1) **Adaptive error feedback for distributed learning**, where A-DEFEAT compensates for communication-compression error in decentralized learning with a new proof technique and convergence analysis. (2) **Learning and Lyapunov-based control for real-world wireless networks**, where objectives are derived from Lyapunov functions that encode real-world constraints, state representations are built from quantities a deployed system can observe, and Q-learning, DDPG, and MADDPG policies are compared in simulation and on live hardware (spectrum sharing on a real-time LoRa testbed, LDPP-MADDPG power allocation, and LYRIC), together with Lyapunov drift-plus-penalty control for multi-hop LoRa adaptation (FORGE-LoRa). (3) **Wireless testbed development**, including the physical-layer stack of the EdgeAI Lab testbed, the UnionLabs remote-access platform, and UAV nodes. (4) **Language-model agents with hardware in the loop**, where SONAR generates network controllers and checks them on physical radios.

Future plans build on these results in three directions. The first is cross-layer learning-based adaptation for heterogeneous networks with safe online learning under non-stationary conditions. The second is digital-twin pipelines calibrated from testbed measurements, with quantitative criteria for when a policy trained in the twin can be deployed over the air. The third is decentralized learning over unreliable wireless networks with time-varying topologies, lossy links,

and quantized communication, connected to secure aggregation under user dropouts. In parallel, I plan to develop UnionLabs into shared community infrastructure federated across institutions, and I would seek external funding through NSF CNS/NeTS and CCF programs, a CAREER proposal, and industry partnerships in AI-for-RAN.

Education

University of Florida, Gainesville, FL, USA 08/2024 – present
Ph.D. Candidate, Electrical and Computer Engineering. Advisor Prof. Mingyue Ji (EdgeAI Lab). Expected completion Spring 2027.
Research. Machine learning for real-world wireless systems; learning-driven network control on live testbeds; decentralized learning.

University of Utah, Salt Lake City, UT, USA 08/2021 – 07/2024
Ph.D. Studies, Electrical and Computer Engineering. Advisor Prof. Mingyue Ji. Transferred to the University of Florida with my advisor's research group in Aug. 2024.

George Washington University, Washington, DC, USA 01/2020 – 06/2021
M.S., Electrical Engineering.

Tianjin University of Science and Technology (TUST), Tianjin, China 09/2015 – 06/2019
B.Eng., Electronic Information Engineering.

Research Experience

Doctoral Researcher, EdgeAI Lab (advisor Prof. Mingyue Ji) 08/2021 – present
University of Utah (08/2021 – 07/2024) and University of Florida (08/2024 – present)

EdgeAI Lab. Real-World Wireless Communication Testbed

University of Florida • Platform for the over-the-air experiments in [C5], [C6], [C8] • Submitted manuscript [C4(S)]

- Implemented the physical-layer software stack of the lab's wireless testbed in C++, covering modulation and demodulation, filtering, synchronization, channel encoding and decoding, and correction of carrier-frequency offset and phase offset.
- Built Python APIs on top of the C++ core so that over-the-air experiments can be scripted, repeated, and parameterized from a single configuration; designed automated experiment workflows that adapt to heterogeneous physical-layer hardware (USRP software-defined radios and LoRa radios).
- Deployed the group's learning and control algorithms (reinforcement-learning spectrum sharing, power allocation, and per-slot LoRa PHY adaptation) on both lab-scale and building-scale deployments of the testbed, and used the platform for the real-time LoRa experiments reported in [C8], the multi-hop LoRa experiments in [C5], and the localization measurements in [C6].
- Worked with srsRAN 4G/5G on USRP software-defined radios in early doctoral projects, alongside protocols implemented on the lab's own physical-layer stack; have recently begun working with OCUDU.
- *Technologies.* C++, Python, USRP software-defined radios, srsRAN 4G/5G, LoRa radios, Linux.

UnionLabs. AWS-Based Remote Access and Sharing of NextG and IoT Testbeds

University of Florida • NSF-funded project (CIRC award, 2025–2027), with collaborators Prof. Zhangyu Guan (Virginia Tech), Dr. Mingzhe Chen (University of Miami), Dr. Bang Chul Jung (Ajou University), Dr. Myeung Suk Oh (Georgia Southern University), and Dr. Kevin Liu (The Ohio State University) • Poster [P4]; demonstrations [P5], [P2]

- Developed and deployed the AWS-hosted web user interface for remote access to the wireless testbeds.
- Designed the physical-layer (PHY) code for different hardware, including USRP software-defined radios and LoRa devices.
- Designed the unified Union API that reads and drives the different PHY implementations and connects them automatically to users' learning or optimization algorithms, and an abstraction layer that lets users run the PHY and Union API with a single command.
- Supported collaborators from several universities in running their learning and optimization algorithms through UnionLabs, including experiments that use multiple testbeds at the same time.
- Developed automatic preparation of the UnionLabs environment and a ready starting point for new users, to be shown as a live demonstration at the Warren B. Nelms Annual IoT Conference in Dec. 2026 [P2].
- The UnionLabs and OSWireless demonstration won **Third Prize** in the Demo Competition of the Warren B. Nelms Annual IoT Conference (Dec. 2025).
- *Technologies.* AWS, Java, JavaScript, ReactJS, JSON, Python, USRP, LoRa.

SONAR. Hardware-in-the-Loop Synthesis of Wireless Systems with Language-Model Agents

University of Florida • Submitted manuscript [C4(S)]

- Built SONAR on NVIDIA's AI Telco Engineer framework so that language-model agents generate executable network controllers, which decide radio settings, routes, and transmissions for a physical radio network.
- Developed the evaluation loop in which recorded-channel simulation screens candidate controllers, selected controllers run on physical radios, and the measurements guide the next design iteration.
- *Technologies.* Python, NVIDIA AI Telco Engineer, language-model agents (Nemotron models via LiteLLM), EdgeAI Lab radio testbed.

Constrained Optimization for Real-World Wireless Problems with Reinforcement Learning

University of Utah and University of Florida • Publications [C8], [C9]; poster [P6]; submitted manuscript [J3(S)] • Collaborators Dr. Arupjyoti Bhuyan (Idaho National Laboratory), Dr. Xiang Zhang (TU Berlin)

- Derived optimization objectives with Lyapunov drift-plus-penalty (LDPP) functions under real-world constraints, turning long-term constrained problems into per-slot decisions that a learning agent can act on.
- Designed novel state representations for reinforcement-learning algorithms applied to communication-system control, and evaluated Q-learning, DDPG, and MADDPG policies both in simulation and in deployment on the EdgeAI Lab testbed.
- Carried out an experimental evaluation of reinforcement learning for spectrum sharing on a real-time LoRa testbed [C8]. Designed and implemented a closed-loop platform that continuously measures received signal strength and interference and feeds them to the agents, so that the agents interact with a real wireless channel; compared centralized and decentralized Q-learning, DDPG, and MADDPG for distributed power allocation under shared spectrum on live hardware.
- Proposed LDPP-MADDPG, a distributed beam-scheduling and power-allocation scheme for mmWave cellular networks in which base stations operate as independent operators over shared spectrum [C9]. The scheme combines a Lyapunov drift-plus-penalty objective with multi-agent DDPG and introduces a local-observation feature that captures the penalty created by the stochastic constraints. An earlier MADDPG power-allocation study used measurement data from the POWDER platform [P6].
- Developed LYRIC, a Lyapunov-constrained multi-agent power-control method evaluated on measured LoRa data [J3(S)].
- *Technologies.* Python, reinforcement learning (Q-learning, DDPG, MADDPG), Lyapunov optimization, LoRa testbed, POWDER measurement data.

Resilient Multi-Hop LoRa Networks. FORGE-LoRa, HYDRA-LoRa, and CONCERT-LoRa

University of Florida • Publication [C5] (IEEE MILCOM 2026); poster [P1]; submitted manuscripts [C3(S)], [J2(S)] • Collaborator Dr. Arupjyoti Bhuyan (Idaho National Laboratory)

- Derived the optimization objective for multi-hop LoRa networks with Lyapunov functions under real-world constraints, maximizing long-term goodput subject to minimum-goodput and average-power constraints.
- Designed the system architecture for jointly adapting the LoRa PHY parameters (spreading factor, coding rate, and transmit power) on a per-slot basis, driven by a packet-error-rate model that maps each link's SNR margin to its expected error rate [C5].
- Building on the joint-adaptation method, designed a path discovery and optimization system that keeps multi-hop communication live against interference and hardware outages.
- Evaluated the controller against the LoRaWAN Adaptive Data Rate (ADR) mechanism in simulation and in deployment on a real multi-hop prototype built from off-the-shelf LoRa hardware on the EdgeAI Lab testbed.
- Extended the LoRa mesh work to anti-jamming operation through HYDRA-LoRa, anti-jamming multi-path selection for LoRa mesh networks [C3(S)], and CONCERT-LoRa, certified concurrency for anti-jamming LoRa mesh networks [J2(S)].
- *Technologies.* LoRa, Python, Lyapunov drift-plus-penalty optimization, multi-hop routing.

LoFiLoc. Device-Free LoRa Fingerprint Indoor Localization with Commodity Radios

University of Florida • Publication [C6] (IEEE MILCOM 2026 Workshops); extension in preparation [C1(P)]

- Developed device-free indoor localization that locates a person who carries no device, using only the received signal strength (RSSI) reported by commodity LoRa radios, so that a link already deployed for communication also senses its environment.
- Designed a session-referenced deviation feature that cancels day-to-day baseline drift, a window-based per-visit

soft vote that suppresses errors from transient interference, and a pluggable classification head with a proposed trust-weighted prototype classifier and an optional random-forest alternative.
- Deployed the system in an occupied seven-room apartment with one transmitter and eight SX1276 LoRa receivers, and evaluated it over calibration sessions collected on multiple days.
- Extending LoFiLoc to LoFiTrack, room-level tracking of a walking person from the RSSI of two time-shared commodity LoRa beacons on one channel, running live and unattended on the same apartment deployment, with calibration from a short guided walk in place of per-point holds; manuscript in preparation for IEEE DySPAN 2027 [C1(P)].
- *Technologies.* LoRa (SX1276), Python, signal-strength fingerprinting, classification.

Transformer-Enhanced DDPG for RIS-Assisted MU-MISO Systems (collaboration)

University of Florida • Publication [C7] (IEEE MILCOM 2026 Workshops)

- Contributed to a transformer-enhanced deep deterministic policy gradient (DDPG) method for the joint design of base-station beamforming and reconfigurable-intelligent-surface (RIS) phase shifts in multi-user MISO systems, with H. Zhang and O. Jiang.

A-DEFEAT. Adaptive Error Feedback for Decentralized Learning

University of Utah and University of Florida • Poster [P3]; submitted manuscript [C2(S)]; journal manuscript in preparation [J1(P)] • Collaborators Dr. Kevin S. Chan and Dr. Jake B. Perazzone (U.S. Army DEVCOM Army Research Laboratory), Prof. Shiqiang Wang (University of Exeter)

- Designed a new decentralized learning algorithm with an adaptive, discounted error-feedback (error-compensation) mechanism for communication compression under data heterogeneity.
- Proposed the key proof technique for the adaptive error-feedback mechanism, established convergence guarantees for the proposed algorithm, and compared its convergence rate with existing algorithms.
- Evaluated the algorithm against existing decentralized learning methods under a range of settings.
- Extending A-DEFEAT and its gradient-tracking variant A-DEFEAT-GT to momentum (A-DEFEAT-MT) and variance-reduction (A-DEFEAT-VR) variants in a unified framework, in a journal manuscript in preparation for *IEEE Transactions on Signal Processing* [J1(P)].
- *Technologies.* Python, optimization theory.

Information-Theoretic Secure Aggregation with Uncoded Groupwise Keys

University of Utah • Publications [J4], [C10] • With K. Wan (Huazhong University of Science and Technology), H. Sun (University of North Texas), M. Ji, and G. Caire (TU Berlin)

- Studied the information-theoretic secure aggregation problem of federated learning, in which the server must learn only the aggregate of the users' models while tolerating user dropouts, under the practical assumption that keys are mutually independent and each key is shared, uncoded, by a group of S users.
- The work characterizes when uncoded groupwise keys are optimal. If $S > K - U$ (with K users and at most $K - U$ dropouts), a new scheme achieves the same optimal communication cost as the best coded-key scheme, whereas if $S \leq K - U$ uncoded groupwise key sharing is strictly sub-optimal.
- Implemented the proposed secure-aggregation scheme on Amazon EC2 and compared it experimentally with existing secure-aggregation schemes that use offline key sharing.

UAV-Based Wireless Communication Platform

University of Florida

- Assembled customized drones and integrated wireless communication payloads for airborne networking experiments; initialized and tuned the flight controllers; designed and mounted the customized communication payloads.
- Designed the communication protocol between ground devices and the airborne node.
- *Technologies.* UAV flight controllers, embedded systems, wireless payload integration.

Master's Projects, George Washington University

- **Age of Information with Energy Harvesting** (advisor Prof. Omur Ozel). Analyzed the age of information (AoI) in M/G/1/1 and idealized infinite-buffer queueing models, including variants with packet deadlines, and investigated AoI-minimization strategies for the bufferless M/G/1/1 model with packet deadlines.
- **Secure File Transfer. Encryption and Decryption for FTP** (advisor Prof. Tian Lan). Designed a custom encryption/decryption scheme for FTP on Linux with client-side encryption and server-side decryption of commands and data, and verified end-to-end encrypted FTP transmission.

Undergraduate Projects, Tianjin University of Science and Technology

- **Voice-controlled smart-home system** (advisor Jing He). Designed STM32F103-based hardware with an LD3320 voice-recognition module; wrote firmware linking the microcontroller to speaker, PC, and LCD peripherals; tested spoken commands with system feedback.
- **License-plate recognition system** (advisor Prof. Yang An). Built a MATLAB pipeline for image capture from surveillance cameras, filtering (Hamming window), and character segmentation and matching.
- **Android music player** (advisor Prof. Ya-Jun Li). Developed a multi-format music player in Java using Eclipse and the Android SDK.
- **Digital counter** (advisor Prof. Yu-liang Liu). Designed the counter circuit, selected components, and assembled, tested, and debugged the board.

Research Collaborators

Government and national laboratories

- Dr. Arupjyoti Bhuyan, Director of the Wireless Security Institute and Directorate Fellow, Idaho National Laboratory (U.S. Department of Energy). Co-author on FORGE-LoRa, HYDRA-LoRa, CONCERT-LoRa, LDPP-MADDPG, and LYRIC.
- Dr. Kevin S. Chan, Lead Research Electronics Engineer, U.S. Army DEVCOM Army Research Laboratory. Co-author on A-DEFEAT.
- Dr. Jake B. Perazzone, U.S. Army DEVCOM Army Research Laboratory. Co-author on A-DEFEAT.

Academic

- Prof. Mingyue Ji, University of Florida (Ph.D. advisor). Co-author on all doctoral work.
- Prof. Zhangyu Guan, Virginia Tech. Collaborator on the NSF-funded UnionLabs project.
- Prof. Shiqiang Wang, University of Exeter. Co-author on A-DEFEAT.
- Prof. Kai Wan (Huazhong University of Science and Technology), Prof. Hua Sun (University of North Texas), and Prof. Giuseppe Caire (Technische Universität Berlin). Co-authors on secure aggregation.
- Dr. Xiang Zhang, Technische Universität Berlin. Co-author on LDPP-MADDPG and LYRIC.
- Dr. Mingzhe Chen, University of Miami. Collaborator on the NSF-funded UnionLabs project.
- Dr. Bang Chul Jung, Ajou University. Collaborator on the NSF-funded UnionLabs project.
- Dr. Myeung Suk Oh, Georgia Southern University. Collaborator on the NSF-funded UnionLabs project.
- Dr. Kevin Liu, The Ohio State University. Collaborator on the NSF-funded UnionLabs project.

Publications

Each list is in reverse chronological order. Entries marked **(S)**, as in [J2(S)], are submitted and under review; **(P)** marks a manuscript in preparation, as in [J1(P)] and [C1(P)]; all other entries are published. My name is shown in bold. Google Scholar profile.

Journal Articles

[J1(P)] **X. Yao**, S. Wang, K. S. Chan, and M. Ji, "A-DEFEAT: A Unified Adaptive Error-Feedback Framework for Compressed Decentralized Learning with Gradient Tracking, Momentum, and Variance Reduction," in preparation for *IEEE Transactions on Signal Processing*.
Journal extension of [C2(S)], generalizing A-DEFEAT and A-DEFEAT-GT to A-DEFEAT-MT and A-DEFEAT-VR.

[J2(S)] **X. Yao**, A. Bhuyan, and M. Ji, "CONCERT-LoRa: Certified Concurrency for Anti-Jamming LoRa Mesh Networks," submitted to *IEEE Internet of Things Journal*.

[J3(S)] **X. Yao**, X. Zhang, A. Bhuyan, and M. Ji, "LYRIC: Lyapunov-Constrained Multi-Agent Power Control with Measured LoRa Evaluation," submitted to *IEEE Transactions on Cognitive Communications and Networking*.

[J4] K. Wan, **X. Yao**, H. Sun, M. Ji, and G. Caire, "On the Information Theoretic Secure Aggregation with Uncoded Groupwise Keys," *IEEE Transactions on Information Theory*, vol. 70, no. 9, pp. 6596–6619, Sept. 2024. DOI 10.1109/TIT.2024.3422087. Preprint arXiv:2204.11364.
Characterizes the optimal communication cost of information-theoretic secure aggregation when keys are mutually independent and each key is shared by a group of S users. Uncoded groupwise keys are optimal when $S > K - U$ and strictly sub-optimal when $S \leq K - U$. Includes an Amazon EC2 implementation compared against existing schemes with offline key sharing.

Conference Papers

[C1(P)] **X. Yao**, A. Bhuyan, and M. Ji, "LoFiTrack: Live Device-Free Tracking of a Walking Person from the RSSI of Two LoRa Beacons on One Channel," in preparation for *IEEE International Symposium on Spectrum Innovation (DySPAN)*, 2027.
Extends [C6] from locating a stationary person under one beacon to room-level tracking of a walking person from two time-shared commodity LoRa beacons on one channel, calibrated from a short guided walk.

[C2(S)] **X. Yao**, S. Wang, J. Perazzone, K. S. Chan, and M. Ji, "Compressed Decentralized Learning with Error-Feedback under Data Heterogeneity" (A-DEFEAT), submitted to *International Conference on Learning Representations (ICLR)*, 2027.

[C3(S)] **X. Yao**, A. Bhuyan, and M. Ji, "HYDRA-LoRa: Anti-Jamming Multi-Path Selection for LoRa Mesh Networks," submitted to *IEEE International Conference on Communications (ICC)*, 2027.

[C4(S)] **X. Yao** and M. Ji, "SONAR: Hardware-in-the-Loop Synthesis of Wireless Systems," submitted to *IEEE International Conference on Communications (ICC)*, 2027.

[C5] **X. Yao**, A. Bhuyan, and M. Ji, "FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks," in *Proc. IEEE Military Communications Conference (MILCOM)*, National Capital Region, USA, Oct. 2026.
Per-slot Lyapunov drift-plus-penalty control that jointly adapts spreading factor, coding rate, and transmit power, combined with path discovery and optimization so that multi-hop LoRa communication stays live under interference and hardware outages; evaluated against LoRaWAN ADR in simulation and on a real multi-hop prototype.

[C6] **X. Yao** and M. Ji, "LoFiLoc: Device-Free LoRa Fingerprint Indoor Localization with Commodity Radios," in *Proc. IEEE Military Communications Conference (MILCOM) Workshops*, National Capital Region, USA, Oct. 2026.
Device-free localization of a person from RSSI alone, reported by commodity LoRa radios, with drift-cancelling features, interference-suppressing voting, and a trust-weighted prototype classifier; deployed in an occupied apartment.

[C7] H. Zhang, O. Jiang, **X. Yao**, and M. Ji, "Transformer-Enhanced DDPG for Beamforming and Phase Shift in RIS-Assisted MU-MISO Systems," in *Proc. IEEE Military Communications Conference (MILCOM) Workshops*, National Capital Region, USA, Oct. 2026.

[C8] **X. Yao**, X. Wang, H. Zhang, and M. Ji, "Experimental Evaluation of Reinforcement Learning for Spectrum Sharing on a Real-Time LoRa Testbed," in *Proc. IEEE International Symposium on Spectrum Innovation (DySPAN) Workshops*, Washington, DC, USA, May 2026, pp. 1–8. DOI 10.1109/DySPAN69846.2026.11571139.
Q-learning, DDPG, and MADDPG spectrum-sharing policies evaluated on live LoRa hardware in real time rather than in simulation only.

[C9] **X. Yao**, A. Bhuyan, X. Zhang, and M. Ji, "A Novel LDPP-MADDPG Approach for Distributed Power Allocation in mmWave Cellular Networks," in *Proc. IEEE Military Communications Conference (MILCOM), 6G and Spectrum Security for Critical Communication (6GSECC) Workshop*, Los Angeles, CA, USA, Oct. 2025. DOI 10.1109/MILCOM64451.2025.11310668.
Distributed power allocation for mmWave cellular networks that combines a Lyapunov drift-plus-penalty (LDPP) objective with multi-agent deep deterministic policy gradient (MADDPG).

[C10] K. Wan, **X. Yao**, H. Sun, M. Ji, and G. Caire, "GroupSecAgg: Information Theoretic Secure Aggregation with Uncoded Groupwise Keys," in *Proc. IEEE International Conference on Communications (ICC)*, Rome, Italy, May–Jun. 2023, pp. 3890–3895. DOI 10.1109/ICC45041.2023.10279171.
Conference version of [J4].

Presentations, Posters and Demonstrations

Conference Presentations

[T1] "FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks," IEEE Military Communications Conference (MILCOM), National Capital Region, USA, Oct. 2026.

[T2] "LoFiLoc: Device-Free LoRa Fingerprint Indoor Localization with Commodity Radios," IEEE MILCOM Workshops, National Capital Region, USA, Oct. 2026.

[T3] "Experimental Evaluation of Reinforcement Learning for Spectrum Sharing on a Real-Time LoRa Testbed," IEEE DySPAN Workshops, Washington, DC, USA, May 2026.

[T4] "A Novel LDPP-MADDPG Approach for Distributed Power Allocation in mmWave Cellular Networks," IEEE MILCOM 6GSECC Workshop, Los Angeles, CA, USA, Oct. 2025.

[T5] "GroupSecAgg: Information Theoretic Secure Aggregation with Uncoded Groupwise Keys," IEEE International Conference on Communications (ICC), Rome, Italy, Jun. 2023.

[T6] "High-Band Frequency Spectrum for 5G" (team presentation), George Washington University, Washington, DC,

USA, 2020.

Posters and Demonstrations

[P1] "FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks" (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2026. (Upcoming.)

[P2] "UnionLabs: Automatic Preparation and a Ready Starting Point for New Users" (live demonstration), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2026. (Upcoming.)

[P3] "A-DEFEAT: An Adaptively Discounted Error-Feedback Algorithm for Decentralized Learning" (poster), North American School of Information Theory (NASIT), Brigham Young University, Provo, UT, USA, Jun. 2026.

[P4] "UnionLabs: Shared Access to Heterogeneous Wireless IoT and Wireless OS Testbeds" (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2025.

[P5] "OSWireless: Automatic System to Maximize Network Throughputs" (live demonstration), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2025. **Third Prize, Demo Competition.**

[P6] "MADDPG-Based Power Allocation Scheme Using Measurement Data from POWDER" (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2024.

Teaching Experience

- **Lab Instructor, EEL 4514 Communication Systems and Components, University of Florida.** Designed and verified the lab sessions on signal processing, held and graded the lab sessions, and held office hours for course-related questions before exams.
- **Teaching Assistant, ECE 5510 Random Processes, University of Utah.** Assisted in designing the lecture slides, graded homework and exams, and held weekly office hours to answer course-related questions.
- **Project Instructor, UnionLabs Web Design (undergraduate project), University of Florida.** Directed and guided students through the web-design part of the UnionLabs project, from background knowledge to design skills including JavaScript, ReactJS, and the communication logic between the web interface and the testbeds.
- **Project Instructor, Automatic System Design (undergraduate project), University of Florida.** Guided students to understand the structure and code of the telco system, an AI-in-the-loop evaluation system, and how to use the code for customized purposes, and introduced hardware-in-the-loop design based on the telco system.
- **Testbed Training and Onboarding, UnionLabs.** Onboard collaborators from multiple universities, ranging from machine-learning researchers new to radios to communications engineers new to reinforcement learning, onto the shared wireless testbeds. Prepare documentation for the customized system, design worked examples, and structure first sessions so that every user runs a real over-the-air experiment quickly.

Mentoring

Undergraduate students, University of Florida

1. Labib M. Hossain, Summer 2026 – Fall 2026. UnionLabs Web Design (JavaScript, ReactJS, and interface-to-testbed communication logic).
2. Grace Mihiral, Fall 2026. Automatic System Design (telco system, AI-in-the-loop evaluation, and hardware-in-the-loop design).
3. Sarthak Gupta, Fall 2026. Automatic System Design (telco system, AI-in-the-loop evaluation, and hardware-in-the-loop design).

Teaching Statement (Summary)

Philosophy. Students learn engineering most effectively when theory is connected to the physical world as early and as often as possible. In wireless communications and machine learning this connection is concrete. A matched filter derived on paper can be observed as a constellation diagram from the student's own transmitter, a coding theorem can be verified as a measured shift in a bit-error-rate curve, and a learning algorithm can be run and measured on real machines. My goal is to help students build a solid conceptual foundation, the practical skill to implement and debug real systems, and the confidence that comes from seeing their own analysis confirmed by measurement.

Methods. Theoretical foundations in class, with emphasis on the mathematical expressions; live software-defined-radio demonstrations so that effects such as intersymbol interference and coding gain are observed as they are derived; project-based homework tested over the air or against recorded real signals through remote testbed access of the kind UnionLabs provides; and projects scaffolded in explicit stages from simulation to hardware, so that students with different preparation all reach the final over-the-air milestone. Learning is assessed through staged milestones that

reward implementation, measurement, and debugging alongside analysis, and the course is adjusted using student feedback throughout the semester.

Mentoring. Early contact with the real system followed by growing ownership. A new student runs a working over-the-air experiment in the first weeks, then takes responsibility for one testbed component or algorithm, growing from reproducing a known result to extending it to leading a first-authored paper, with regular individual meetings, projects matched to each student's strengths, and frequent presentations.

Courses I am prepared to teach. Signals and systems, digital communications, and digital signal processing (undergraduate); wireless system design and networking courses that connect the physical and MAC layers; machine learning for wireless networks (graduate); and a new project-based course, *Hands-On Wireless Systems with Software-Defined Radios*, in which students implement a complete communication link and then apply a learning algorithm to control it.

Professional Service

- Reviewer for *IEEE Transactions on Information Theory* (TIT).
- Reviewer for *IEEE Transactions on Parallel and Distributed Systems* (TPDS).
- Reviewer for *IEEE Transactions on Green Communications and Networking* (TGCN).
- Reviewer for *IEEE Transactions on Cognitive Communications and Networking* (TCCN).
- Reviewer for *IEEE Communications Magazine*.
- Reviewer for *Computer Networks* (Elsevier).
- Reviewer for *EURASIP Journal on Wireless Communications and Networking* (JWCN).
- Judge, Gator Quant Hack Competition, 2026.
- Student Volunteer, IEEE Military Communications Conference (MILCOM), National Capital Region, USA, 2026.
- Student Volunteer, IEEE International Conference on Computer Communications (INFOCOM), 2022.

Technical Skills

- **Programming.** Python, C/C++, Java, JavaScript, ReactJS, MATLAB, Arduino, JSON, XHTML/DHTML.
- **Machine learning and AI.** Reinforcement learning (Q-learning, DDPG, MADDPG, transformer-enhanced policies), federated learning, decentralized learning, communication compression and error feedback, secure aggregation, distributed and parallel computing, language-model agents for controller synthesis (NVIDIA AI Telco Engineer).
- **Wireless communications and signal processing.** Physical-layer implementation (modulation and demodulation, synchronization, filtering, channel encoding and decoding, carrier-frequency and phase-offset correction), software-defined radio (USRP), srsRAN 4G/5G, LoRa and low-power multi-hop networking, mmWave power allocation, RIS-assisted systems, real-world testbed development, network protocol analysis, signal-strength fingerprinting for localization.
- **Optimization and theory.** Lyapunov drift-plus-penalty optimization, convergence analysis of decentralized learning, information-theoretic secure aggregation, queueing models and age of information.
- **Embedded, cloud, and systems.** STM32 microcontrollers, Android SDK (Eclipse/JDK), AWS (EC2 and web deployment), Linux, UAV flight-controller setup and payload integration.
- **Design and productivity tools.** AutoCAD, Siemens NX, Sharp3D, Adobe (Photoshop, Premiere Pro, After Effects, Acrobat), MS Office, LaTeX.

Honors and Awards

- Overall Runner-Up (Second Prize), NSF SpectrumX Student Data & Algorithm Competition (SpX-DAC), team with X. Wang and Y. Hu, advised by Prof. Mingyue Ji; 37 teams from 25 institutions. 2026
- Third Prize, Demo Competition, Warren B. Nelms Annual IoT Conference, University of Florida (UnionLabs / OSWireless demonstration). 2025
- Third-Prize Scholarship, Tianjin University of Science and Technology. 2017
- Advanced Individual of Social Work, Tianjin University of Science and Technology. 2016
- Excellent League Member, Tianjin University of Science and Technology. 2016

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